臺中榮民總醫院

鼻咽癌治療指引

Nasopharyngeal Cancer Treatment Guideline –Taichung Veterans General Hospital

Version 01. 2020

本指引僅供參考，最終仍需主治醫師、病人及家屬共同決定

目錄

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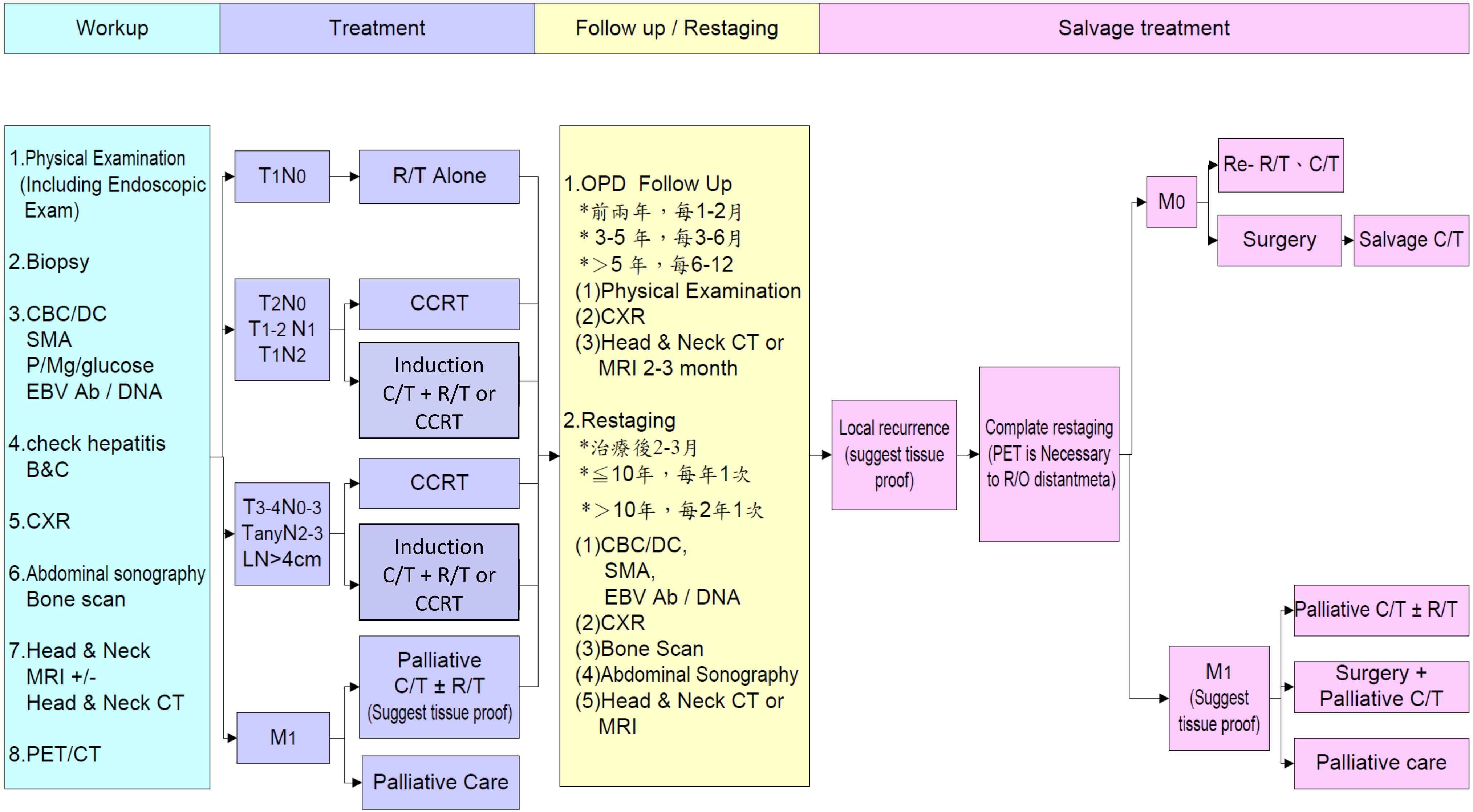
流程表 (FLOW CHART)



今年更新內容 (WHAT'S NEW)

- Concurrent chemotherapy 新增 Regimen 6: Oral form chemotherapy

指引內文 (CONTENT)

1. 目的(Goal)：制定鼻咽癌放射治療指引與執行規範。

2. 適用範圍(Application)：適用於根治性鼻咽癌病人之放射治療。

3. 放射治療之適應症(Indications for radiotherapy)

Staging	Treatment	Adjuvant treatment
T1N0M0	1. Radiotherapy alone	Nil
T1-4N1-3M0 T2N0M0 T1-2N1M0 T3-4N0-3M0 TanyN2-3M0 LN ≥ 4cm	1. Neoadjuvant chemotherapy followed by radiotherapy or Concurrentchemoradiotherapy (CCRT) 2. CCRT	Suggest adjuvant oral chemotherapy for patients with T4N(+) or N2 and LN>4cm or N3 disease

放射治療指引 (RADIATION TECHNIQUES)

1. 模擬定位(Simulation)

(1) 病人治療時需採端正仰躺姿勢，下巴微上揚，雙肩盡量下垂，雙上肢對稱擺於身體兩側。

(2) 放射治療師依上述姿勢為病人量身訂做製作治療固定模具(面具)。

(3) 在電腦斷層掃描定位室，請病患依原姿勢躺好後戴上固定模具。

(4) 在病患之模具或皮膚上，貼上金屬標記，在影像上呈現中心點之位置。

(5) 電腦斷層之掃描範圍及條件為由腦部至肺尖應包含眼眶、鼻咽、頸部，切片厚度 3-4 毫米。治療之掃描範圍及切片厚度條件，可由主治醫師自行訂定。

(6) 透過靜脈注射顯影劑，可加強判讀腫瘤侵犯之範圍，但如果病患腎功能差或有其他不適應症則除外。

(7) 其他影像：可考慮安排磁振造影(MRI)，或正子，若病人接受 induction chemotherapy 後，則視其需要安排電腦斷層掃描影像(PET/CT)，作為分期、腫瘤圈選輔助使用。

2. 治療體積定義(Target Volume Definition)及放射治療計畫規劃(Radiation Therapy Planning)

(1) GTV (Gross tumor volume)：鼻咽部腫瘤、頸部轉移淋巴結。由 CT 影像、臨床訊息、PET/CT 或 MRI 影像可判讀之主要腫瘤(primary tumor)及臨床上呈陽性之淋巴腺(直徑大於 1 公分之腫瘤或含有壞死區域之腫瘤)。如果放射治療前做過化學治療時，需考慮化學治療前腫瘤之侵犯範圍。

(2) CTV (Clinical tumor volume)：包含 GTV 及腫瘤可能侵犯之範圍，可分為 CTV1 及 CTV2。

A. CTV1：包含原發腫瘤的 GTV 加上 0.4~1 公分之範圍與兩側上頸部淋巴結(level II~III)。如為 N3 病人或下頸部淋巴結呈陽性，則應包含同側 level III–V 區域之頸部淋巴結。

B. CTV2：為 CTV1 加上 0.4~1 公分之範圍。

(3) PTV (Planning Target Volume)：考慮擺位誤差及內部器官移動，放射治療計畫在 CTV2 周圍至少要加上 3~5 毫米之安全範圍來形成 PTV。如果 GTV 或 CTV 太靠近腦幹或脊索時，安全範圍可縮小至 1 毫米。

3. 治療劑量(Radiation dose)

(1) CTV1: 一般給予劑量為一天一次治療，每次劑量 200 雷德，共 35 次或 33 次，總劑量 7000 或 6600 cGy。 如因腫瘤侵犯至腦部或太過於接近腦幹時，可考慮 hyperfractionated radiotherapy: 一天兩次治療，每次劑量 120 雷德，共 7680 雷德/64 次治療。

(2) CTV2: 一般給予劑量 6000 cGy/ 35 次治療或 5940 cGy/ 33 次。

(3) 預防性照射: 兩側頸部 level III-IV 可以給予放射治療 4500~5000 雷德/25 次治療。

(4) 治療天數應由病況決定，合理範圍為 45~56 天。

(5) 治療次數與劑量仍需依照臨床醫師判斷，以上僅供參考。

4. 危急器官定義之劑量限制

(1) 腦幹(Brain stem)：最高劑量 < 54 Gy。

(2) 脊索(Spinal cord)：最高劑量 < 45 Gy。

(3) 下頜骨(Mandible)、顳下頜關節(TMJ)：最高劑量<70Gy。

(4) 聲帶(Glottic Larynx)：平均劑量< 45 Gy。

(5) 唾液腺(Parotid Glands): 劑量限制為至少符合以下其中一個條件：

A. 平均劑量< 26 Gy(至少有一側之唾液腺符合)。

B. 至少兩側唾液腺 20 cc 之體積接受< 20 Gy。

C. 中值劑量(median dose) < 30 Gy (至少一側唾液腺符合)。

5. 治療驗證(Treatment Verification)

(1) 三度空間放射治療或強度調控放射治療：治療前由放射師拍攝正交之驗證片(orthogonal verification films)來驗證照野中心點的位置是否正確。

(2) 影像導引放射治療(IGRT)：如需要影像導引放射治療，治療前應由放射師拍攝電腦斷層影像或正交之驗證片來確認治療範圍，治療期間應由醫師決定再次驗證的時間及頻率。

6. 放射治療日程(Radiotherapy schedule)

療程第一天開始放射線治療，一天一次，週一至周五，通常 33-35 次，約 7 週。

化學治療指引 (PRINCIPLES OF SYSTEMIC THERAPY)

1. Concurrent chemotherapy

(1) 配方一(Regimen 1): Q4W [Cisplatin 20 mg/m2 + 5-FU 400 mg/m2, Day 1-4], for 2 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	20 mg/m2	D1-4	IVD	Normal saline
5-FU	400 mg/m2	D1-4	IVD	Normal saline

(2) 配方二(Regimen 2): Q4W[Cisplatin 80 mg/m2 Day1+5-FU 500 mg/m2, Day1-3], for 2 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	80 mg/m2	D1	IVD	Normal saline
5-FU	500 mg/m2	D1-4	IVD	Normal saline

(3) 配方三(Regimen 3): QW [Cisplatin 30~50 mg/m2], for 6-8 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	30-50 mg/m2	D1	IVD	Normal saline

(4) 配方四(Regimen 4): Q3W [Cisplatin 100 mg/m2], for 2-3 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	100 mg/m2	D1	IVD	Normal saline

(5) 配方五(Regimen 5): Cetuximab (Erbitux)

※須搭配化學藥物使用，並視病人情況及臨床醫師判斷決定是否使用。

Loading dose 400 mg/m2 and maintenance dose 250 mg/m2

藥品	劑量	途徑	稀釋溶液
Cetuximab (loading dose)	400 mg/m2	IVD	no
Cetuximab (maintenance dose)	250 mg/m2	IVD	no

(6) 配方六(Regimen 6): Oral form chemotherapy

UFUR 100mg (Tegafur 100mg, Uracil 224mg) 1 tab 1# TID during R/T

(7) 其他配方(Other regimens): according to the physician

2. Adjuvant chemotherapy

(1) 配方一(Regimen 1): Ufur 1 TAB PO TID

藥品	劑量	途徑	頻 次
Ufur	1 TAB	po	TID

(2) 配方二(Regimen 2): Ufur 2 TAB PO BID + Endoxan 50mg 1 TAB PO QD

藥品	劑量	途徑	頻 次
Ufur	2 TAB	po	BID
Endoxan 50mg	1 TAB	po	QD

3. Induction chemotherapy

(1) **配方一(Regimen 1):** Q2W [Cisplatin 60mg/m2 on Day 1, 5-FU 2500mg/m2 + leucovorin 250mg/m2 on Day 8], for 5-6 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	60 mg/m2	D1	IVA	Normal saline
5-FU	2500 mg/m2	D8	IVA	Normal saline
Leucovorin	250 mg/m2	D8	IVA	Normal saline

(2) **配方二(Regimen 2):** Q3W [Cisplatin 100mg/m2 on Day 1, 5-FU 1000mg/m2 on Day 1-4], for 2-3 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	100 mg/m2	D1	IVA	Normal saline
5-FU	1000 mg/m2	D1-4	IVA	Normal saline

(3) **配方三(Regimen 3):** Q3W [Mitomycin 8mg/m2 + Epirubin 60mg/m2 + Cisplatin 60mg/m2], for 2-4 courses

藥品	劑量	天數	途徑	稀釋溶液
Mitomycin	8 mg/m2	D1	IVA	Normal saline
Epirubicin	60 mg/m2	D1	IVA	Normal saline
Cisplatin	60 mg/m2	D1	IVA	Normal saline

(4) **配方四(Regimen 4):** Q4W [Cisplatin 60mg/m2 on Day 1, Gemzar 1000mg/m2 on Day 8, 5-FU 2500mg/m2 + leucovorin 250mg/m2 on Day15, and Vinorelbine 25mg/m2 on Day 22], for 3 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	60 mg/m2	D1	IVA	Normal saline
Gemzar	1000 mg/m2	D8	IVA	Normal saline
5-FU	2500 mg/m2	D15	IVA	Normal saline
Leucovorin	250 mg/m2	D15	IVA	Normal saline
Vinorelbine	250 mg/m2	D22	IVA	Normal saline

(5) **配方五(Regimen 5):** Q4W [Gemzar 1000 mg/m2 on Day 1,8,15 and Cisplatin 60 mg/m2 on Day 22]

藥品	劑量	天數	途徑	稀釋溶液
Gemzar	1000 mg/m2	D1, 8, 15	IVA	Normal saline
Cisplatin	60 mg/m2	D22	IVA	Normal saline

(6) 配方六(Regimen 6): Cetuximab (Erbitux)

※須搭配化學藥物使用，並視病人情況及臨床醫師判斷決定是否使用。

Loading dose 400 mg/m2 and maintenance dose 250 mg/m2

藥品	劑量	途徑	稀釋溶液
Cetuximab (loading dose)	400 mg/m2	IVD	no
Cetuximab (maintenance dose)	250 mg/m2	IVD	no

(7) 其他配方(Other regimens): according to the physician

4. Salvage chemotherapy

(1) 配方一(Regimen 1): Q4W [Cisplatin 20 mg/m2 + 5-FU 400 mg/m2, Day 1-4], for 2 courses

藥品	劑量	天數	途徑	稀釋溶液
Cisplatin	20 mg/m2	D1-4	IVD	Normal saline
5-FU	400 mg/m2	D1-4	IVD	Normal saline

(2) 配方二(Regimen 2): Q3W [Mitomycin 8mg/m2 + Epirubin 60mg/m2 + Cisplatin 60mg/m2], for 2-4 courses

藥品	劑量	天數	途徑	稀釋溶液
Mitomycin	8 mg/m2	D1	IVA	Normal saline
Epirubicin	60 mg/m2	D1	IVA	Normal saline
Cisplatin	60 mg/m2	D1	IVA	Normal saline

(3) 配方三(Regimen 3): Q4W [Gemzar 1000 mg/m2 on Day 1,8,15 and Cisplatin 60 mg/m2 on Day 22]

藥品	劑量	天數	途徑	稀釋溶液
Gemzar	1000 mg/m2	D1,8,15	IVA	Normal saline
Cisplatin	60 mg/m2	D22	IVA	Normal saline

(4) 配方四(Regimen 4): Q4W [Gemzar 1000 mg/m2 on Day 1,8,15 and Vinorelbine 25 mg/m2 on Day 22]

藥品	劑量	天數	途徑	稀釋溶液
Gemzar	1000 mg/m2	D1,8,15	IVA	Normal saline
Vinorelbine	25 mg/m2	D22	IVA	Normal saline

(5) 配方五(Regimen 5): Cetuximab (Erbitux)

※須搭配化學藥物使用，並視病人情況及臨床醫師判斷決定是否使用。

Loading dose 400 mg/m2 and maintenance dose 250 mg/m2

藥品	劑量	途徑	稀釋溶液
Cetuximab (loading dose)	400 mg/m2	IVD	no
Cetuximab (maintenance dose)	250 mg/m2	IVD	no

(6) 其他配方(Other regimens): According to the physician

PS: If the patient has poor renal function, we may shift cisplatin to carboplatin.

※ **<u>以上藥物均配合健保給付有所調整，並依病人情況及臨床醫師判斷決定使用方式</u>**。

腫瘤分期 (STAGING)

American Joint Committee on Cancer (AJCC)
TNM Staging System for the Nasopharynx (8th ed., 2017)
(The following types of cancer are not included: Mucosal melanoma, lymphoma, sarcoma of the soft tissue, bone and cartilage.)

Primary Tumor (T)

TX Primary tumor cannot be assessed

T0 No tumor identified, but EBV-positive cervical node(s) involvement

Tis Carcinoma *in situ*

T1 Tumor confined to nasopharynx, or extension to oropharynx and/or nasal cavity without parapharyngeal involvement

T2 Tumor with extension to parapharyngeal space, and/or adjacent soft tissue involvement (medial pterygoid, lateral pterygoid, prevertebral muscles)

T3 Tumor with infiltration of bony structures at skull base, cervical vertebra, pterygoid structures, and/or paranasal sinuses

T4 Tumor with intracranial extension, involvement of cranial nerves, hypopharynx, orbit, parotid gland, and/ or extensive soft tissue infiltration beyond the lateral surface of the lateral pterygoid muscle

Regional Lymph Nodes (N)

NX Regional lymph nodes cannot be assessed

N0 No regional lymph node metastasis

N1 Unilateral metastasis in cervical lymph node(s) and/or unilateral or bilateral metastasis in retropharyngeal lymph node(s), 6 cm or smaller in greatest dimension, above the caudal border of cricoid cartilage

N2 Bilateral metastasis in cervical lymph node(s), 6 cm or smaller in greatest dimension, above the caudal border of cricoid cartilage

N3 Unilateral or bilateral metastasis in cervical lymph node(s), larger than 6 cm in greatest dimension, and/or extension below the caudal border of cricoid cartilage

Distant Metastasis (M)

M0 No distant metastasis

M1 Distant metastasis

Histologic Grade (G)

A grading system is not used for NPCs.

Anatomic Stage/Prognostic Groups

Stage 0	Tis	N0	M0
Stage I	T1	N0	M0
Stage II	T0, T1	N1	M0
	T2	N0, N1	M0
Stage III	T0, T1, T2	N2	M0
	T3	N0, N1, N2	M0
Stage IVA	T4	N0, N1, N2	M0
	Any T	N3	M0
Stage IVB	Any T	Any N	M1

參考文獻 (REFERENCE)

1. NCCN guidelines in oncology-v.1.2019
2. Wu Q, Manning M, Schmidt-Ullrich R, Mohan R. The potential for sparing ofparotids and escalation of biologically effective dose with intensity-modulatedradiation treatments of head and neck cancers: a treatment design study. Int JRadiatOncolBiol Phys 2000;46(1):195-205.
3. Dogan N, King S, Emami B, et al. Assessment of different IMRT boost deliverymethods on target coverage and normal-tissue sparing. Int J RadiatOncolBiolPhys 2003;57(5):1480-1491.
4. Mohan R, Wu Q, Morris M, et al. “Simultaneous Integrated Boost” (SIB) IMRT ofadvanced head and neck squamous cell carcinomas—dosimetric analysis. Int JRadiatOncolBiol Phys 2001;51(3):180–181.
5. M. Johnston, S. Clifford, R. Bromley, M. Back, L. Oliver, T. Eade. Volumetric-modulated arc therapy in head and neck radiotherapy: a planning comparison using simultaneous integrated boost fornasopharynxand oropharynx carcinoma.ClinOncol 2011:23(8):503-11
6. Lin JC, Jan JS, Hsu CY, et al. Phase III study of concurrent chemoradiotherapy versus radiotherapy alone for advanced nasopharyngeal carcinoma: positive effect on overall and progression-free survival. J ClinOncol. 2003 Feb 15;21(4):631-7
7. Chan AT, Leung SF, Ngan RK, et al. Overall survival after concurrent cisplatin-radiotherapy compared with radiotherapy alone in locoregionally advanced nasopharyngeal carcinoma. J Natl Cancer Inst 2005;97:536-539.
8. Chitapanarux I, Lorvidhaya V, Kamnerdsupaphon P, et al. Chemoradiation comparing cisplatin versus carboplatin in locally advanced nasopharyngeal cancer: Randomised, non-inferiority, open trial. Eur J Cancer 2007;43:1399-1406.
9. Al-Sarraf M, LeBlanc M, Giri PG, et al. Chemoradiotherapy versus radiotherapy in patients with advanced nasopharyngeal cancer: phase III randomized Intergroup study 0099. J ClinOncol 1998;16:1310-1317.
10. Lee NY, Harris J, Garden A, et al. Phase II multi-institutional study of IMRT ± chemotherapy for nasopharyngeal carcinoma (RTOG 0225): Preliminary results. Int J RadiatOncolBiol Phys 2007;69:S13-S14.
11. Wee J, Tan EH, Tai BC, et al. Randomized trial of radiotherapy versus concurrent chemoradiotherapy followed by adjuvant chemotherapy in patients with American Joint Committee on Cancer/International Union Against Cancer Stage III and IV nasopharyngeal cancer of the endemic variety. J ClinOncol 2005;23:6730-6738.
12. Chih-Wen Twu,Wen-Yi Wang, Chien-Chih Chen, et al.Metronomic Adjuvant Chemotherapy Improves TreatmentOutcome in Nasopharyngeal Carcinoma Patients WithPostradiation Persistently Detectable Plasma Epstein-BarrVirus Deoxyribonucleic Acid.Int J Radiation OncolBiol Phys, 2014; 89, 21-29.
13. Lin JC, Jan JS, Chen KY, et al. Outpatient weekly 24-hour infusional adjuvant chemotherapy of cisplatin, 5-fluorouracil, and leucovorin for high-risk nasopharyngeal carcinoma. Head Neck. 2003 Jun;25(6):438-50
14. Lin JC, Jan JS, Hsu CY, et al. Outpatient weekly neoadjuvant chemotherapy followed by radiotherapy for advanced nasopharyngeal carcinoma: high complete response and low toxicity rates. Br J Cancer. 2003 Jan 27;88(2):187-94
15. Vermorken JB, Mesia R, Rivera F, et al. Platinum-based chemotherapy plus cetuximab in head and neck cancer.N Engl J Med 2008;359:1116-1127.
16. Zhang L, Zhang Y, Huang PY, et al. Phase II clinical study of gemcitabine in the treatment of patients with advanced nasopharyngeal carcinoma after the failure of platinum-based chemotherapy. Cancer ChemotherPharmacol 2008;61(1):33-38. Epub 2007 Mar 20.

17.Vermorken JB, Trigo J, Hitt R, et al. Open-label, uncontrolled, multicenter phase II study to evaluate the efficacy and toxicity of cetuximab as a single agent in patients with recurrent and/or metastatic squamous cell carcinoma of the head and neck who failed to respond to platinum-based therapy. J Clin Oncol 2007;25(16):2171-2177.
18.Zhang L, Zhang Y, Huang PY, et al. Phase II clinical study of gemcitabine in the treatment of patients with advanced nasopharyngeal carcinoma after the failure of platinum-based chemotherapy. Cancer ChemotherPharmacol 2008;61(1):33-38. Epub 2007 Mar 20.
19.Chan AT, Hsu MM, Goh BC, et al. Multicenter, phase II study of cetuximab in combination with carboplatin in patients with recurrent or metastatic nasopharyngeal carcinoma. J ClinOncol 2005;23:3568-3576.
20.Chih-Wen Twu, Wen-Yi Wang, Chien-Chih Chen, et al. Metronomic Adjuvant Chemotherapy Improves TreatmentOutcome in Nasopharyngeal Carcinoma Patients WithPostradiation Persistently Detectable Plasma Epstein-BarrVirus Deoxyribonucleic Acid.Int J Radiation OncolBiol Phys, 2014; 89, 21-29.
21.Lin JC, Jan JS, Chen KY, et al. Outpatient weekly 24-hour infusional adjuvant chemotherapy of cisplatin, 5-fluorouracil, and leucovorin for high-risk nasopharyngeal carcinoma. Head Neck. 2003 Jun;25(6):438-50
22.Lin JC, Jan JS, Hsu CY, et al. Outpatient weekly neoadjuvant chemotherapy followed by radiotherapy for advanced nasopharyngeal carcinoma: high complete response and low toxicity rates. Br J Cancer. 2003 Jan 27;88(2):187-94
23.Vermorken JB, Mesia R, Rivera F, et al. Platinum-based chemotherapy plus cetuximab in head and neck cancer. N Engl J Med 2008;359:1116-1127.
24.Zhang L, Zhang Y, Huang PY, et al. Phase II clinical study of gemcitabine in the treatment of patients with advanced nasopharyngeal carcinoma after the failure of platinum-based chemotherapy. Cancer ChemotherPharmacol 2008;61(1):33-38. Epub 2007 Mar 20.
25.Vermorken JB, Trigo J, Hitt R, et al. Open-label, uncontrolled, multicenter phase II study to evaluate the efficacy and toxicity of cetuximab as a single agent in patients with recurrent and/or metastatic squamous cell carcinoma of the head and neck who failed to respond to platinum-based therapy. J Clin Oncol 2007;25(16):2171-2177.
26.Zhang L, Zhang Y, Huang PY, et al. Phase II clinical study of gemcitabine in the treatment of patients with advanced nasopharyngeal carcinoma after the failure of platinum-based chemotherapy. Cancer ChemotherPharmacol 2008;61(1):33-38. Epub 2007 Mar 20.
27. Chan AT, Hsu MM, Goh BC, et al. Multicenter, phase II study of cetuximab in combination with carboplatin in patients with recurrent or metastatic nasopharyngeal carcinoma. J ClinOncol 2005;23:3568-3576.
28.Chih-Wen Twu, Wen-Yi Wang, Chien-Chih Chen, et al.Metronomic Adjuvant Chemotherapy Improves TreatmentOutcome in Nasopharyngeal Carcinoma Patients WithPostradiation Persistently Detectable Plasma Epstein-BarrVirus Deoxyribonucleic Acid.Int J Radiation OncolBiol Phys, 2014; 89, 21-29.